In[ ]:=
```
η = 1
α2 = 0.197 ;
α2bar = 0.197;
N1 = 1;
g1 = 3;
N2 = 1 ;
g2 = 1;
N3 = 1 ;
g2bar = 1;
NSolve[{Q2bar == -((6 * η) / (5 * (5 * η - 1) * α2bar)) (g1 * (β12bar)^(3/2) + g2 (β22bar)^(3/2)),
  Q2bar == -((6 * η) / (5 * (5 * η - 1) * β12bar)) g2bar -
    (9 * η) / (10 * (5 * η - 1) * β12bar) (((N1 g1 - 1) (β12bar)^(3/2)) / N1 + g2 (β22bar)^(3/2)),
  Q2bar == -((6 * η) / (5 * (5 * η - 1) * α2 * β22bar)) g2bar - (9 η) / (10 (5 η - 1) α2 β22bar)
    (((N2 g2 - 1) (β22bar)^(3/2)) / N2 + g1 (β12bar)^(3/2))}, {β12bar, β22bar, Q2bar}]

NSolve[{Q1 == (-9 η) / (10 (5 η - 1)) (N1 g1 - 1) / N1}, {Q1}]

NSolve[{Q2 == (-9 η) / (10 (5 η - 1) γ12) ((N1 g1 - 1) / N1 (γ12)^(3/2) + g2),
  Q2 == (-9 η) / (10 (5 η - 1) α2) ((N2 g2 - 1) / N2 + g1 (γ12)^(3/2))}, {Q2, γ12}]
```

Out[ ]= 1

Out[ ]=
```
{{β12bar → -0.456301476 + 0.535395506 i, β22bar → 1.29222184 + 0.131956951 i,
  Q2bar → 0.36720419 + 0.384790185 i}, {β12bar → -0.456301476 - 0.535395506 i,
  β22bar → 1.29222184 - 0.131956951 i, Q2bar → 0.36720419 - 0.384790185 i},
 {β12bar → -0.0220495409 + 0.122341852 i, β22bar → -0.803554801 + 0.498417028 i,
  Q2bar → 1.20988142 + 0.842974899 i}, {β12bar → -0.0220495409 - 0.122341852 i,
  β22bar → -0.803554801 - 0.498417028 i, Q2bar → 1.20988142 - 0.842974899 i},
 {β12bar → 0.275303609, β22bar → 0.962157373, Q2bar → -2.09714746}}
```

Out[ ]= {{Q1 → -0.45}}

Out[ ]=
```
{{Q2 → -0.851982695, γ12 → 0.395423204},
 {Q2 → 0.48683122 + 0.266046503 i, γ12 → -0.224554347 + 0.194487676 i},
 {Q2 → 0.48683122 - 0.266046503 i, γ12 → -0.224554347 - 0.194487676 i}}
```

In[ ]:=
```
η = 2
α2 = 0.197;
α2bar = 0.197;
N1 = 2;
g1 = 3;
N2 = 1;
g2 = 2;
N2bar = 1;
g2bar = 2;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1)}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1) (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]
NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) ((N1 g1 - 1)/N1 (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) ((N2 g2 - 1)/N2 + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]
```

Out[ ]= 2

Out[ ]= {{Q1 → -0.5}}

Out[ ]= {{γ12bar → 0.497450802, Q2bar → -1.42478379}}

Out[ ]= {{β12 → 0.248575024, β22bar → 0.882615033, Q2 → -3.63754985}}

```
η = 3
α2 = 0.197;
α2bar = 0.197;
N1 = 3;
g1 = 3;
N2 = 3;
g2 = 1;
N2bar = 3;
```

```
g2bar = 1;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1 (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]
NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) ((N1 g1 - 1)/N1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) ((N2 g2 - 1)/N2 + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]

η = 4
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 3;
g1 = 4;
N2 = 2;
g2 = 2;
N2bar = 2 ;
g2bar = 2 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1 (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]
NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
```

$$Q2 == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\beta 12}\,g2bar\,(\beta 22bar)^{\frac{3}{2}} - \frac{9\,\eta}{10\,(5\,\eta - 1)\,\beta 12}\left(\frac{(N1\,g1 - 1)}{N1}\,(\beta 12)^{\frac{3}{2}} + g2\right),$$

$$Q2 == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\alpha 2}\,g2bar\,(\beta 22bar)^{\frac{3}{2}} - \frac{9\,\eta}{10\,(5\,\eta - 1)\,\alpha 2}\left(\frac{(N2\,g2 - 1)}{N2} + g1\,(\beta 12)^{\frac{3}{2}}\right)\Big\},$$

$$\{\beta 12,\ \beta 22bar,\ Q2\},\ Reals\Big]$$

```
η = 5
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 5 ;
g1 = 3 ;
N2 = 5 ;
g2 = 1 ;
N2bar = 5 ;
g2bar = 1 ;
```

$$NSolve\Big[\Big\{Q1 == \frac{-9\ \eta}{10\,(5\,\eta - 1)}\,\frac{(N1\,g1 - 1)}{N1}\Big\},\ \{Q1\},\ Reals\Big]$$

$$NSolve\Big[\Big\{Q2bar == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\alpha 2bar}\,g1\,(\gamma 12bar)^{\frac{3}{2}},\ Q2bar == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\gamma 12bar}\,g2bar - \frac{9\ \eta}{10\,(5\,\eta - 1)}\,\frac{(N1\,g1 - 1)}{N1}\,(\gamma 12bar)^{\frac{1}{2}}\Big\},\ \{\gamma 12bar,\ Q2bar\},\ Reals\Big]$$

$$NSolve\Big[\Big\{Q2 == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\alpha 2bar\,\beta 22bar}\left(g1\,(\beta 12)^{\frac{3}{2}} + g2\right),$$

$$Q2 == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\beta 12}\,g2bar\,(\beta 22bar)^{\frac{3}{2}} - \frac{9\,\eta}{10\,(5\,\eta - 1)\,\beta 12}\left(\frac{(N1\,g1 - 1)}{N1}\,(\beta 12)^{\frac{3}{2}} + g2\right),$$

$$Q2 == \frac{-6\,\eta}{5\,(5\,\eta - 1)\,\alpha 2}\,g2bar\,(\beta 22bar)^{\frac{3}{2}} - \frac{9\,\eta}{10\,(5\,\eta - 1)\,\alpha 2}\left(\frac{(N2\,g2 - 1)}{N2} + g1\,(\beta 12)^{\frac{3}{2}}\right)\Big\},$$

$$\{\beta 12,\ \beta 22bar,\ Q2\},\ Reals\Big]$$

```
η = 6
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 6 ;
g1 = 3 ;
N2 = 3 ;
```

```
g2 = 2 ;
N2bar = 3 ;
g2bar = 2 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1)}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1) (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]

NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) (((N1 g1 - 1)/N1) (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) (((N2 g2 - 1)/N2) + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]

η = 7
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 7 ;
g1 = 3 ;
N2 = 7 ;
g2 = 1 ;
N2bar = 7 ;
g2bar = 1 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1)}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1) (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]
```

```
NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) ((N1 g1 - 1)/N1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) ((N2 g2 - 1)/N2 + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]

η = 8
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 8 ;
g1 = 3 ;
N2 = 4 ;
g2 = 2 ;
N2bar = 4 ;
g2bar = 2 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1 (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]

NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) ((N1 g1 - 1)/N1 (β12)^(3/2) + g2),
   Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) ((N2 g2 - 1)/N2 + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]

η = 9
```

```
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 9 ;
g1 = 3 ;
N2 = 9 ;
g2 = 1 ;
N2bar = 9 ;
g2bar = 1 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1}, {Q1}, Reals]
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1 (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]

NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) ((N1 g1 - 1)/N1 (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) ((N2 g2 - 1)/N2 + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]


η = 10
α2 = 0.197 ;
α2bar = 0.197 ;
N1 = 10 ;
g1 = 3 ;
N2 = 5 ;
g2 = 2 ;
N2bar = 5 ;
g2bar = 2 ;
NSolve[{Q1 == (-9 η)/(10 (5 η - 1)) (N1 g1 - 1)/N1}, {Q1}, Reals]
```

```
NSolve[{Q2bar == (-6 η)/(5 (5 η - 1) α2bar) g1 (γ12bar)^(3/2), Q2bar ==
    (-6 η)/(5 (5 η - 1) γ12bar) g2bar - (9 η)/(10 (5 η - 1)) ((N1 g1 - 1)/N1) (γ12bar)^(1/2)}, {γ12bar, Q2bar}, Reals]

NSolve[{Q2 == (-6 η)/(5 (5 η - 1) α2bar β22bar) (g1 (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) β12) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) β12) (((N1 g1 - 1)/N1) (β12)^(3/2) + g2),
  Q2 == (-6 η)/(5 (5 η - 1) α2) g2bar (β22bar)^(3/2) - (9 η)/(10 (5 η - 1) α2) (((N2 g2 - 1)/N2) + g1 (β12)^(3/2))},
 {β12, β22bar, Q2}, Reals]
```

Out[ ]= 3

Out[ ]= {{Q1 → -0.514285714}}

Out[ ]= {{γ12bar → 0.395423204, Q2bar → -0.973694508}}

Out[ ]= {{β12 → 0.225596, β22bar → 0.859355921, Q2 → -2.00718359}}

Out[ ]= 4

Out[ ]= {{Q1 → -0.694736842}}

Out[ ]= {{γ12bar → 0.455668266, Q2bar → -1.57780954}}

Out[ ]= {{β12 → 0.220467444, β22bar → 0.82371058, Q2 → -3.75834918}}

Out[ ]= 5

Out[ ]= {{Q1 → -0.525}}

Out[ ]= {{γ12bar → 0.398720621, Q2bar → -0.958513168}}

Out[ ]= {{β12 → 0.213937891, β22bar → 0.824988804, Q2 → -1.9948906}}

Out[ ]= 6

Out[ ]= {{Q1 → -0.527586207}}

Out[ ]= {{γ12bar → 0.505242179, Q2bar → -1.35780976}}

Out[ ]= {{β12 → 0.213211812, β22bar → 0.794688088, Q2 → -3.64016137}}

Out[ ]= 7

Out[ ]= {{Q1 → −0.529411765}}

Out[ ]= {{γ12bar → 0.400144239, Q2bar → −0.95231418}}

Out[ ]= {{β12 → 0.209024895, β22bar → 0.810493443, Q2 → −1.99094765}}

Out[ ]= 8

Out[ ]= {{Q1 → −0.530769231}}

Out[ ]= {{γ12bar → 0.506227299, Q2bar → −1.35014371}}

Out[ ]= {{β12 → 0.20906053, β22bar → 0.784133789, Q2 → −3.64394797}}

Out[ ]= 9

Out[ ]= {{Q1 → −0.531818182}}

Out[ ]= {{γ12bar → 0.400937847, Q2bar → −0.948946414}}

Out[ ]= {{β12 → 0.206319638, β22bar → 0.802505711, Q2 → −1.98909437}}

Out[ ]= 10

Out[ ]= {{Q1 → −0.532653061}}

Out[ ]= {{γ12bar → 0.506819569, Q2bar → −1.34561326}}

Out[ ]= {{β12 → 0.206600775, β22bar → 0.777852493, Q2 → −3.64656724}}